

Homology-Based Variant-Effect Predictors Break Down on Cytochrome P450 Pharmacogenes

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Supplementary materials accompanying the main text.

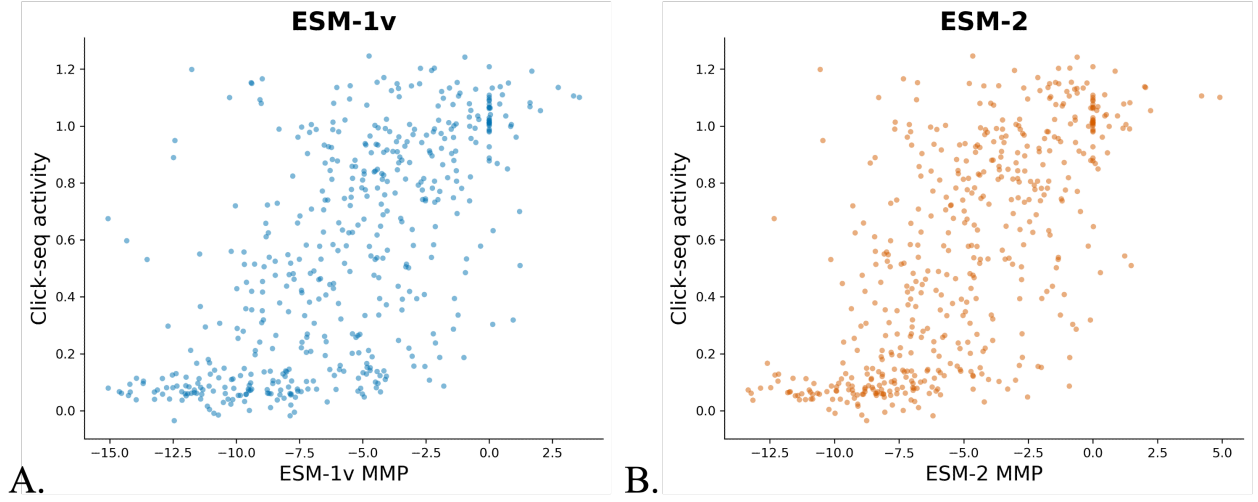


Fig. S1. ESM-1v vs ESM-2 masked marginal probability (MMP) as a predictor of CYP2C9 activity. ESM-1v (**A**) and ESM-2 (**B**) MMP against normalized CYP2C9 DMS click-seq activity for a shared set of 500 randomly sampled variants. Two-sided Spearman's rank correlation: ESM-1v $\rho = 0.657$ ($p = 3.721 \times 10^{-63}$); ESM-2 $\rho = 0.697$ ($p = 5.282 \times 10^{-74}$). ESM-2 correlates with DMS activity more strongly than ESM-1v, a significant difference by a Steiger–Meng test for dependent correlations ($z = -2.92$, two-sided $p = 0.0035$). On this basis ESM-2 was adopted as the masked marginal predictor throughout.

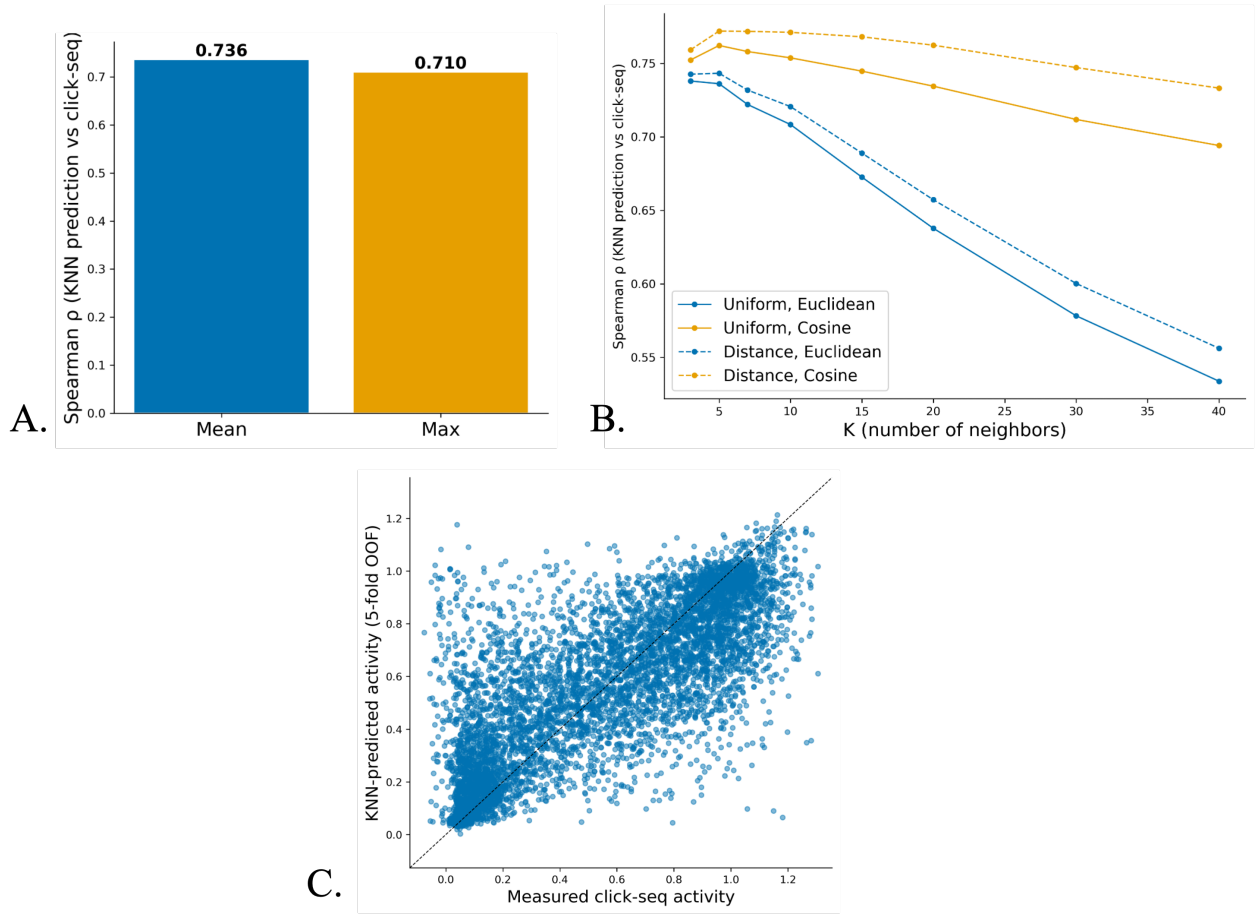


Fig. S2. KNN hyperparameter selection. Cross-validated performance of the ESM-2 embedding KNN regressor predicting normalized CYP2C9 click-seq activity, measured as Spearman correlation between 5-fold out-of-fold (OOF) predictions and measured activity over $N=6,527$ variants. Hyperparameters were tuned in two stages. **A.** Stage 1 compared mean versus max pooling of the final-layer ESM-2 embeddings under default settings ($K=5$, uniform weights, Euclidean distance); mean pooling ($\rho = 0.736$) outperformed max pooling ($\rho = 0.710$). **B.** Holding mean pooling fixed, Stage 2 swept neighbor weighting (uniform vs. distance-weighted), distance metric (Euclidean vs. cosine), and number of neighbors $K \in \{3, 5, 7, 10, 15, 20, 30, 40\}$ jointly, shown as Spearman ρ (y-axis) against K (x-axis) for each weighting $\times$ metric combination. **C.** OOF predicted activity versus measured click-seq activity, with the dashed line indicating perfect prediction ($y = x$). The selected configuration (mean pooling, cosine distance, distance-weighted, $K=5$) reached $\rho = 0.801$ ($p < 10^{-16}$) and was used in all subsequent analyses.

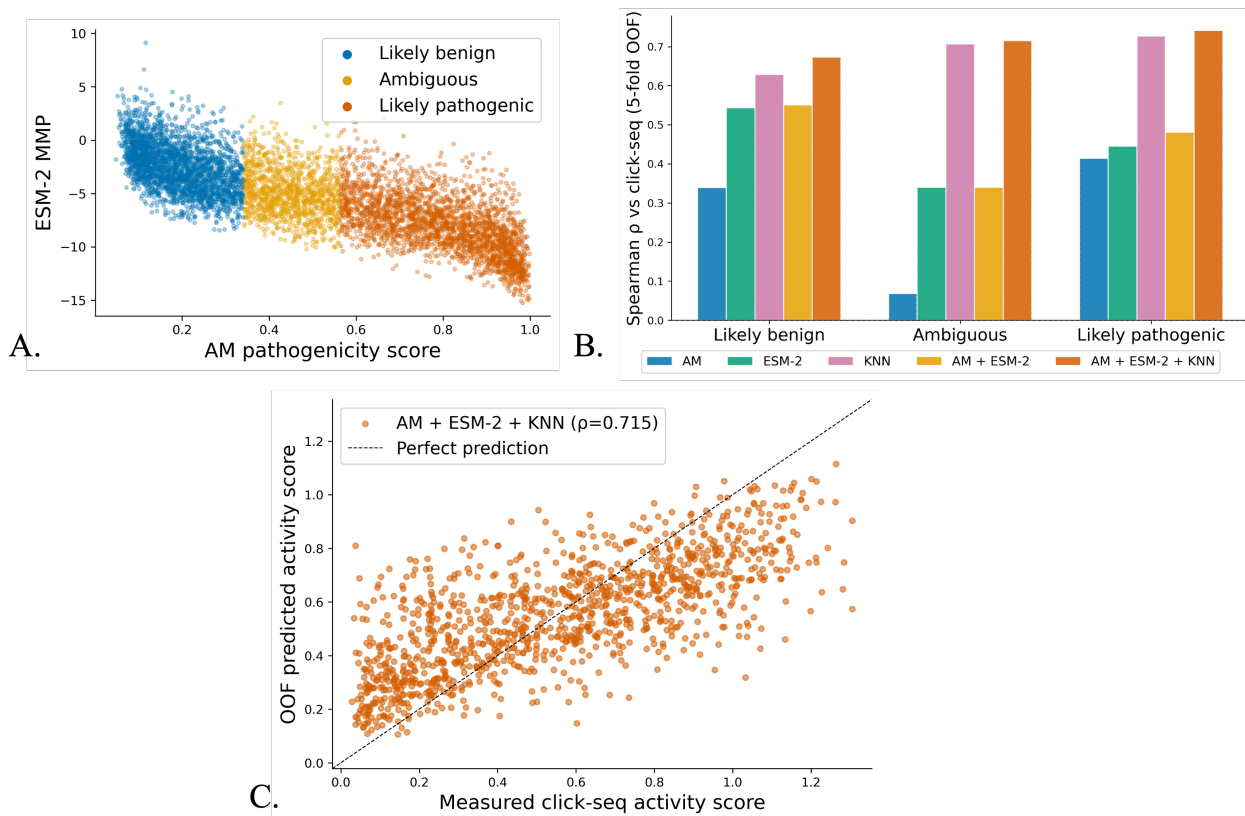


Fig. S3. AM-ESM-2 relationship and ensemble performance on CYP2C9 activity: **A.** ESM-2 MMP versus AM pathogenicity score for CYP2C9 variants, with points colored by AM class (likely benign, ambiguous, likely pathogenic); pooled $\rho = -0.797$. **B.** Spearman correlation with click-seq activity (5-fold out-of-fold) within each AM class for five models: AM only, ESM-2 only, KNN only, AM + ESM-2, and AM + ESM-2 + KNN. **C.** Out-of-fold predicted activity versus measured click-seq activity for ambiguous-class variants, shown for the ensemble model AM + ESM-2 + KNN with ρ annotated; the dashed line indicates $y = x$.

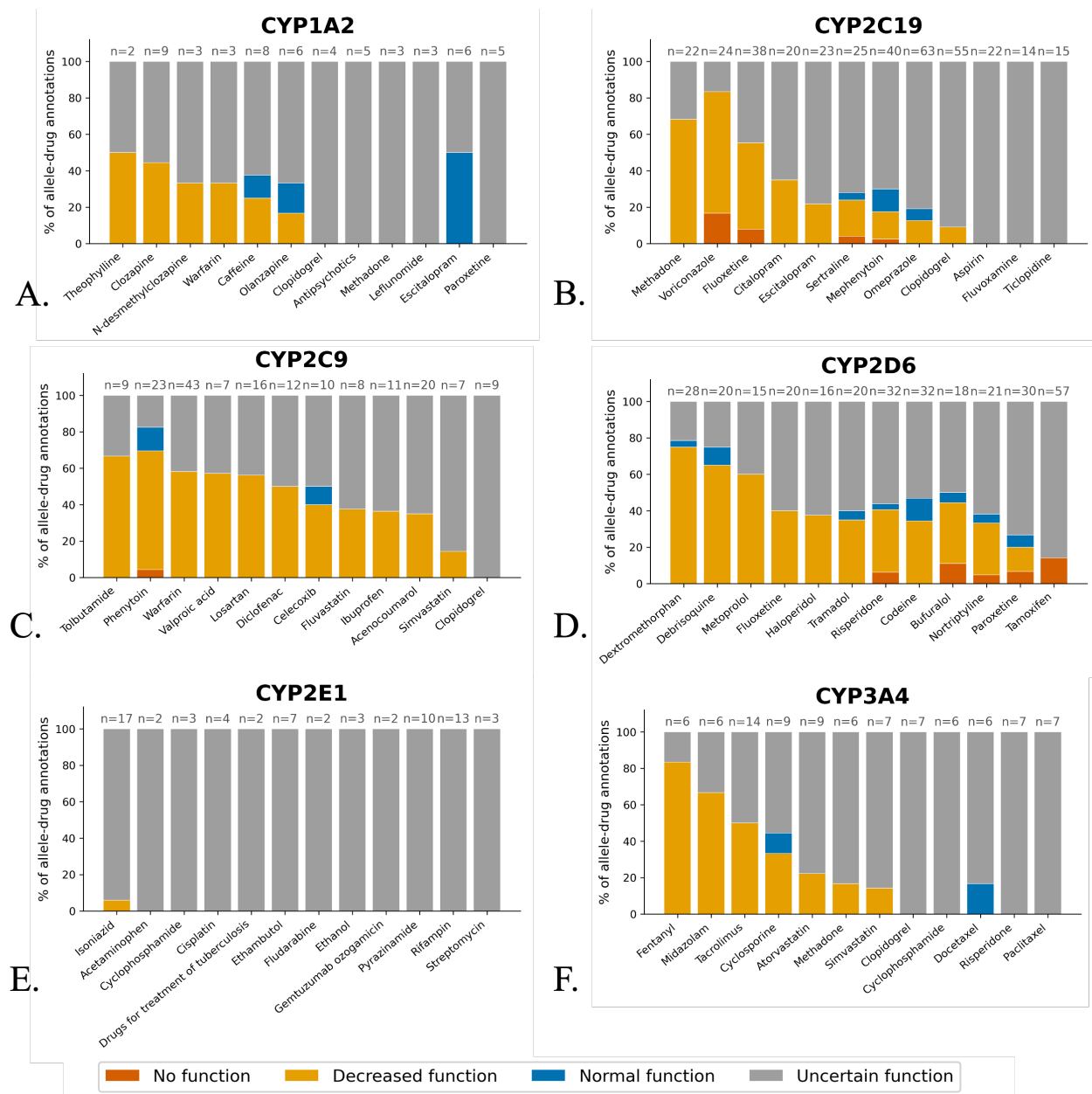


Fig. S4. Substrate-resolved variant effects across six CYPs from ClinPGx. Stacked bar charts showing, for each of the six CYP genes (CYP1A2, CYP2C19, CYP2C9, CYP2D6, CYP2E1, CYP3A4; **A-F**), the distribution of ClinPGx function labels (no function, decreased function, normal function, and uncertain function) across drug substrates (x-axis). Each bar is one drug and shows the percentage of allele-drug annotations assigned each function label (y-axis, % of allele-drug annotations, 0–100%), where one annotation is a single allele-drug pair. The number of annotations (n) is given above each bar. This is the six-gene extension of Fig. 5.